

Generic Identifiability and Directed Containment for Strongly Tree-Child Level-2 Networks under K2P

Principal positive domain theorem and reproducibility manifest

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Promotion status

This is the promoted theorem manuscript for the principal positive K2P domain. Every analytic, graph-theoretic, raw-four-port, five-port, restoration, cycle, ordinary-triangle, probe, genericity, reconstruction, continuous-time, and weak-sharpness layer cited below has passed its stated independent replay and adversarial mutations. In particular, the corrected all-primitive one- and two-port package has zero unresolved rows, zero missing or multiple parent bindings, zero cycles, and zero incoherent transport restrictions. Its exact primary, independent-replay, mutation, and ledger hashes are frozen in Section 7 and Appendix A. The promotion guard passes.

Theorems 8.3, 9.1, 10.1, 11.1, and 12.1 are therefore unconditional under their stated hypotheses. “Unconditional” here means that no computational gate remains; it does not remove any domain, regularity, network-class, or genericity hypothesis written in the theorem statements.

Every statement is confined to the domain defined below. Nothing here classifies any other stochastic sign component.

1. Networks, model, and relations

Let X be a finite labelled leaf set. A *standard semi-directed network* is obtained from a rooted binary phylogenetic network by suppressing the root, forgetting directions on ordinary arcs, and retaining the arrowheads entering reticulations, under the standard no-parallel-edge and suppression conventions used by the primitive graph enumeration. It is *level 2* when each blob has cyclomatic number at most two. It is *strongly tree-child* when every admissible rooted representative is tree-child. These are the conventions used throughout; in particular, “strong” is not being replaced by existence of one tree-child rooting.

For the group $\mathbb{Z}_2 \times \mathbb{Z}_2$, use character order $0, C, G, T$. A K2P edge has Fourier spectrum

$$(1, s, g, s).$$

The parameter domain in the main theorem is

$$\mathcal{D}_+ = \{(s, g) : 0 < s < 1, 0 < g < 1, g > 2s - 1\}. \quad (1.1)$$

Every inheritance probability is in $(0, 1)$. For a network N , let $\Theta_+(N)$ be the corresponding open physical parameter space, let Φ_N be its Fourier polynomial map, let $\mathcal{M}_+(N) = \Phi_N(\Theta_+(N))$, and put

$$d_N = \max_{\theta \in \Theta_+(N)} \text{rank } D\Phi_N(\theta).$$

A source point is *regular* when its Jacobian rank is d_N .

We use a source-relative, section-bearing notion of containment. Write $N \preceq_+ N'$ when there are a regular point $\theta_0 \in \Theta_+(N)$, a connected open neighborhood $U \subseteq \Theta_+(N)$ on which Φ_N has rank d_N , and a real analytic map $\sigma : U \rightarrow \Theta_+(N')$ such that

$$\Phi_N = \Phi_{N'} \circ \sigma \quad \text{on } U. \quad (1.2)$$

Write $N \bowtie_+ N'$ when the two images contain a common analytic germ which is regular and full-dimensional in both images, with physical analytic sections from both parameter spaces. These relations concern regular image germs. They do not mean equality or inclusion of the complete stochastic images.

An *ordinary triangle redirection* changes only the semi-directed orientation of a three-cycle factor incident with exactly three external arms. Write $N \equiv_\Delta N'$ when their labelled decorated trees of blobs agree, corresponding non-triangle factors are related by labelled mixed-graph isomorphisms, and each remaining corresponding factor differs by an ordinary triangle redirection, with all boundary transports coherent. This is a structural quotient; it is not a claim that every redirected representative contains every particular distribution of every other representative.

2. Domain, subdivision, and root invariance

Lemma 2.1 (physical domain and subdivision)

The strict stochastic K2P inequalities for $(1, s, g, s)$ are

$$1 - g > 0, \quad 1 + 2s + g > 0, \quad 1 - 2s + g > 0.$$

On the positive-eigenvalue component these reduce to (1.1). Every point of $\mathcal{D}_+$ admits a strict K2P subdivision. In particular, for all sufficiently small $\varepsilon > 0$,

$$(s_B, g_B) = (1 - \varepsilon, 1 - \varepsilon), \quad (s_A, g_A) = \left(\frac{s}{1 - \varepsilon}, \frac{g}{1 - \varepsilon} \right)$$

are both strict physical pairs and multiply coordinatewise to (s, g) .

Proof. The inequalities are the inverse Fourier probabilities. Their strictness is open, and substitution in the displayed factorization gives the required sufficiently small interval for ε . This avoids the invalid assertion that arbitrary coordinatewise square roots are physical. $\square$

Lemma 2.2 (root movement, including a retained arrowhead)

For the symmetric K2P model, Φ_N is unchanged by moving the root to any admissible rooting of the same standard semi-directed graph.

Proof. Along an ordinary root-to-leaf path, reverse only ordinary arcs; K2P transition matrices are symmetric. Subdivide the new root edge using Lemma 2.1 and suppress the old root. If the old root lies on an edge entering a reticulation, inspect each switching. When that parent is selected, the two old root arms occur only through their serial products. When the other parent is selected, the remaining one-child root stem subtends every observed leaf and therefore carries total Fourier character zero and multiplier one. Every displayed-tree summand is unchanged. Strong tree-childness supplies the required admissible tree-child rooting. $\square$

Thus $\preceq_+$, $\bowtie_+$, and the structural quotient are well-defined for the standard semi-directed objects rather than for a chosen root presentation.

3. Pointwise recovery of the decorated tree of blobs

Lemma 3.1 (displayed-quartet separation)

If two strict physical K2P restrictions display different quartet sets, their Fourier tensors are unequal. More precisely, after a physical port permutation, one of

$$q_{GGGG} - q_{GGTT}, \quad q_{GGGG} - q_{GGTT} - q_{GTTG} + q_{GTGT} \quad (3.1)$$

is zero on one restriction and strictly positive on the other. Positive inheritance mixtures preserve the strict sign.

The graph-theoretic displayed-quartet theorem therefore reconstructs the labelled tree of blobs pointwise from every tensor in $\mathcal{M}_+(N)$. In particular, $N \preceq_+ N'$ implies equality of the labelled trees of blobs.

Lemma 3.2 (degree-three decoration)

For each physical orientation of a three-boundary restriction, the whole-map polynomial

$$\mathcal{T}_i = V^2 X_g - X_s^2 Y_g Z_g \quad (3.2)$$

vanishes identically on the ordinary degree-three tree factor and has a fixed strict nonzero sign on the corresponding three-cycle factor throughout $\mathcal{D}_+$. Hence the ordinary-vertex/three-cycle decoration is also pointwise determined. A strong theta factor has at least four physical boundaries and cannot be confused with an ordinary three-boundary factor.

Lemmas 3.1 and 3.2 use whole semi-directed maps. No rooted restriction label is used as an oracle.

4. The complete two-sector bridge fibre

Cut every bridge of the common decorated tree of blobs. In each nonzero character block, positivity and rank-one uniqueness leave an incidence scale at each side. The two K2P character orbits are $\{C, T\}$ and $\{G\}$.

Lemma 4.1 (equality of the C - and T -scales)

Let a positive normalized local boundary tensor be transformed by positive incidence functions $c_e(h)$, with $c_e(0) = 1$, and suppose both the original and transformed tensors are K2P-invariant under the exchange $C \leftrightarrow T$. Put

$$\rho_e = \frac{c_e(C)}{c_e(T)} > 0.$$

If the component is marked by a retained physical block, compare the positive conservation-supported entries with C, C , respectively T, T , at incidence e and that block, with zero elsewhere. Invariance before and after scaling gives $\rho_e = 1$ for every incidence.

If the component is unmarked and has degree $d \geq 3$, the analogous comparison at every pair $e \neq f$ gives

$$\rho_e \rho_f = 1.$$

Three distinct incidences and positivity force all $\rho_e = 1$.

For $d = 2$, and only there, the argument leaves the genuine stabilizer

$$(\rho_1, \rho_2) = (t, t^{-1}), \quad t > 0. \quad (4.1)$$

This case does not occur in the retained class. The graph-derived primitive supports have respectively 3 ports for the cycle, 4 ports for $\theta_0, \theta_1, \theta_3$, and 5 ports for θ_2 ; restoration and probing only add marked ports; and every unmarked ordinary component has degree at least three. The only simple reduced two-boundary theta has path lengths $(1, 2, 2)$, namely $K_4 - e$. Its exact graph replay enumerates 25 rooted binary acyclic presentations and zero tree-child presentations, so it is excluded from the strong class.

Theorem 4.2 (complete bridge fibre)

On $\mathcal{D}_+$, the complete positive fibre created by cutting and regluing bridges is exactly

$$P_v \mapsto P_v \prod_{e \ni v} a_{v,e}^{1[h_e \in \{C,T\}]} b_{v,e}^{1[h_e = G]}, \quad (4.2)$$

$$s_e \mapsto \frac{s_e}{a_{u,e} a_{v,e}}, \quad g_e \mapsto \frac{g_e}{b_{u,e} b_{v,e}}. \quad (4.3)$$

There is no additional positive gauge. The action is free on all retained components and has no two-sector holonomy.

Proof. All-zero normalization fixes the zero-character scale. Positive rank-one uniqueness gives one scale in each character block. Lemma 4.1 identifies the C - and T -scales, while the singleton G -scale is independent. For an unmarked degree- d component, the pair-anchor exponent rows

$$(12), (13), (23), (14), \dots, (1d)$$

have rank d , with leading determinant -2 , separately in the paired and singleton sectors. Marked components use physical identity anchors. These give analytic normalizers and freeness. Peeling the bridge tree excludes any holonomy. $\square$

Lemma 4.3 (physical local product saturation)

The quotient chart of Theorem 4.2 restricts to a physical local product chart; it is not merely an ambient positive-tensor quotient.

Proof. For an effective bridge (s, g) , choose $r < 1$, sufficiently close to one, such that

$$(r, r), \quad (s/r^2, g/r^2), \quad (r, r) \quad (4.4)$$

all lie in the interior of $\mathcal{D}_+$. Around this interior serial split, the two endpoint pairs vary in four independent coordinates while the residual pair remains physical. Their ratios realize the four incidence directions in (4.2)–(4.3). Suppression of the two serial bivalent vertices returns the same standard bridge. Make these choices independently on the bridge tree and shrink to a constant-rank chart. $\square$

No step asserts that an arbitrary normalized local tensor is itself a physical network tensor.

5. Physical localization, marginal openness, and restoration

Lemma 5.1 (paired marginal open image)

For every $m \geq 1$, the serial-edge map

$$((s_1, g_1), \dots, (s_m, g_m)) \mapsto \left(\prod_i s_i, \prod_i g_i \right) \quad (5.1)$$

maps $\mathcal{D}_+^m$ onto $\mathcal{D}_+$, has differential rank two, and has a local physical analytic section.

Proof. Given $(S, G) \in \mathcal{D}_+$, choose r with

$$\max\{S, G, 2S - G, 0\} < r^{m-1} < 1. \quad (5.2)$$

Use (r, r) on the first $m - 1$ factors and $(S/r^{m-1}, G/r^{m-1})$ on the last. The inequalities in (5.2) put the last pair strictly in $\mathcal{D}_+$. Varying (S, G) gives the local section and rank two. $\square$

By Lemmas 2.2, 4.3, and 5.1, a hypothetical global containment localizes to each corresponding component and to every graph-derived rigid support. Incoming structural boundaries can be selected by root movement, and omitted physical roles can be restored by open marginals.

Lemma 5.2 (fixed-full restoration implication)

Fix actual networks N, N' , a full relation (1.2), a particular omitted leaf label, a source insertion edge, a target attachment, and the induced boundary transport. Marginalizing that same full relation produces the enumerated child relation, and Lemma 5.1 makes its source restriction open. Therefore any exact separator for every such physical child excludes the fixed full parent relation.

The quantifiers in this lemma are essential: it neither lifts an abstract selected relation nor inverts a target deletion map. A graph-terminal relation with an omitted role is expanded relation-first and is not silently reassigned to the obstruction forest.

6. Frozen bounded finite theorem

All rows in this section are generated from primitive graph encodings, retain the source-to-target direction and all physical port data, and bind exactly one evidence object. The authoritative ledgers contain no rooted classification fields.

6.1 Four-port theta universe

The six four-port theta sources yield

$$6(831 + 1983)4! = 405,216$$

directed raw presentations, with no duplicate or missing raw identifier:

displayed-quartet exclusion	360,408	(6.1)
whole-map $\mathcal{T}_i$ strict-sign exclusion	16,974	
exact directed rank exclusion	23,822	
direct terminal presentation	1,472	
restoration-member presentation	2,540.	

The terminal presentations bind to 934 exact terminal classes. The restoration members bind one-to-one to membership records for exactly 997 canonical parent obligations. Each rank row contains a nonzero exact source minor and a symbolic target upper certificate, with strict source-lower greater than target-upper. Every quartet row binds its exact displayed quartet witness.

The compact terminal registry partitions the 934 classes into 839 exact multihomogeneous quadratics, 36 exact direct polynomial separators, four additional F2/F3/F4 hard-case bindings, 20 labelled mixed-graph isomorphisms, and 35 ordinary-triangle quotients. The 36-record direct polynomial overlay is exactly 22 quintic port-orbit obstructions, 12 quartic obstructions, and two cubic obstructions. Each has identically zero target pullback, nonzero rational source pullback, and a strict physical source witness. The independent compiler replay and mutation suite bind the record-to-certificate assignment and reject cubic, quartic, or quintic reassignment.

6.2 Five-port θ_2 universe

The four minimum-repaired θ_2 sources yield

$$4(1983 + 4155)5! = 2,946,240$$

directed presentations:

displayed-quartet exclusion	2,942,592	
whole-map $\mathcal{T}_i$ strict-sign exclusion	2,528	
exact directed rank exclusion	800	(6.2)
direct quadratic separator	240	
labelled isomorphism	80.	

All 2,528 sign rows replay on the whole maps. The 56 dummy-bearing isomorphism roots generate 864 physical descendants with exactly one parent and coherent restricted transport. The terminating descendant forest has 832 leaves: 760 displayed-quartet exclusions and 72 labelled isomorphisms; it has no missing continuation layer, cycle, or unresolved row.

6.3 Three-port cycle universe

The authoritative base ledger has 13,440 directions:

whole-map $\mathcal{T}_i$ strict-sign exclusion	7,452	
fixed-full restoration obligation	5,964	
labelled isomorphism	8	(6.3)
ordinary triangle relation	16.	

The 5,964 roots generate 536,364 physical completions, partitioned as

displayed-quartet strict separator	535,920	
whole-map $\mathcal{T}_i$ strict-sign exclusion	300	
exact directed quadratic separator	132	(6.4)
labelled isomorphism	12.	

There are zero childless roots, unresolved rows, incoherent transports, or legacy rooted reason fields.

6.4 The 997 restoration obligations

The corrected fixed-full forest has 36,568 first children. Of these, 35,758 have displayed-quartet separators, 606 have whole-map $\mathcal{T}_i$ strict-sign certificates, 148 have exact multihomogeneous quadratics, 24 have transported exact quartics, and 32 are continuations. Those 32 parents have 256 second children: 248 quartet separators and eight whole-map $\mathcal{T}_i$ separators. Thus the forest has 36,824 edges, 36,792 terminal leaves, maximum depth two, zero cycles, zero missing children, zero unresolved leaves, and coherent transport restriction on every edge. Every first child has exact graph relation **none**, so an omitted-role graph terminal has not been smuggled into the obstruction forest. Lemma 5.2 therefore discharges all 997 full-parent obligations.

The exact hashes of these frozen ledgers and their independent replays are listed in Appendix A.

7. Coherent one- and two-port completion

The frozen probe-input contract contains 176 distinct physical anchor paths: 43 four-port direct/restored anchors, 96 restored θ_2 anchors, 36 cycle anchors, and one ordinary-tree anchor. Its exact relation census is 143 labelled isomorphisms and 33 ordinary triangles. Every edge of each suppressed mixed graph is an insertion site, including pendant arms, reticulation-incoming edges, and the root-suppressed segment. Artificial root halves are identified only after exact semi-directed isomorphism.

Closed gate G_{probe}

The final corrected probe package is frozen with the following values:

Primary certificate file SHA-256

6c228e0d495fbc68b1df4ddb571558425a35e081cc22b954f232f8a3458991d3

Primary payload SHA-256

674853fa730c4f54b9ba264d539a51591c8b926ad444195e68df086c26f83825

Independent replay file SHA-256

879a397a165e19c162561435e7996a5de5ebdf5e9a7dc38af61a13cc6aa36410

Independent replay payload SHA-256

65160636abfa33de47136a222081ac70bd7b6fae0e029b7a7c379e2d8653df74

Mutation report file SHA-256

b0df0584163150c9a823b4e364b8ee46c196ae8abb28fdca4d3d5893a97bfea7

Mutation report payload SHA-256

58006ed7b6677c055b5cdd7249857dc2f752fb3db9cfbcc5bbe5e0a26e31875f

Combined ordered-ledger root

7868fed6f8e0c10fcb9740da8ffdc7f64ea68939c99cba6f364da4cfd90bf50

$$\text{One-port directed rows: } 29,964 \quad \text{Two-port directed rows: } 544,571 \quad (7.1)$$

The combined root is the SHA-256 of the canonical object containing the one-port, two-port-parent-inventory, and two-port ordered roots, keyed by those three names. The exact census is

$$\begin{aligned} 29,964 &= 27,758 \text{ quartet} + 99 \mathcal{T}_i + 1,915 \text{ isomorphism} + 192 \text{ triangle}, \\ 544,571 &= 511,266 \text{ quartet} + 576 \mathcal{T}_i + 30,969 \text{ isomorphism} + 1,760 \text{ triangle}. \end{aligned} \quad (7.2)$$

The 2,107 one-port equalities form 469 exact graph-pair-plus-transport classes and generate every two-port parent exactly once. All 32,729 two-port equalities have their reversed one-port marginal in that universe. The 33 triangle anchors generate exactly 192 one-port and 1,760 two-port triangle equalities, all transporting the same inherited ordinary triangle; no new triangle is created. Every required zero gate is zero.

The machine placeholder additionally binds the exact byte hashes of the one-port ledger, two-port parent inventory, two-port ledger, exact transport ledger, parent-restriction ledger, and separation-proof registry. Promotion requires independent primitive reconstruction of all 176 anchors, complete site Cartesian coverage, exact-relation precedence, exact quartet and whole-map $\mathcal{T}_i$ replay, transport and parent-restriction replay, reverse-order coverage, global-triangle coherence, and targeted mutations.

Lemma 7.1 (word reconstruction)

For two surviving physical local factors, one-port restrictions determine the directed core segment containing each omitted boundary and two-port restrictions determine the order of every pair on a common segment. Hence the complete labelled subdivision word on every segment is determined. All certified transports assemble to one labelled mixed-graph isomorphism, except for one coherent orientation choice at each ordinary triangle.

Proof. Root at the fixed incoming support port and suppress only unported bivalent vertices. Every additional boundary attaches to a unique subdivision of a finite directed core segment. Its one-port restriction records that segment. A two-port restriction of two boundaries on the same segment records their order. Because these comparisons arise from actual words they are transitive and recover the unique total word. The probe gate binds every child transport to its parent, so the reconstructed segment words agree on overlaps. A probe through a triangle edge fixes a literal orientation; otherwise the unique ordinary triangle survives with its stored edge set. The choices are therefore coherent. $\square$

Theorem 7.2 (bounded local relation)

For every pair H, H' of retained physical primitive factors,

$$H \preceq_+ H' \iff H \cong H' \text{ as labelled mixed graphs} \quad \text{or} \quad H \equiv_{\Delta} H'. \quad (7.3)$$

Proof. The ledgers in Section 6 assign every raw direction exactly once to a strict pointwise separator, a strict rank exclusion, an exact direct algebraic separator, a restoration obligation, a labelled isomorphism, or an ordinary triangle. Lemma 5.2 and the terminating forests discharge every restoration obstruction. Gate G_{probe} and Lemma 7.1 assemble all equality terminals and exclude every incoherent full extension. No raw direction remains. Conversely, labelled isomorphisms transport parameters, and ordinary triangles have the common-germ construction in Section 8. $\square$

8. Global directed-containment classification

Lemma 8.1 (necessity)

Let N, N' be binary standard semi-directed strongly tree-child level-2 networks on the same labelled leaf set. If $N \preceq_+ N'$, then $N \equiv_{\Delta} N'$.

Proof. Section 3 forces the same labelled decorated tree of blobs. Theorems 4.2 and Lemma 4.3 extract corresponding local factors, with exactly the two incidence gauges and no hidden holonomy. Lemmas 2.2 and 5.1 transfer each localized relation to a rigid physical support. Theorem 7.2 classifies every such direction. The one-/two-port transports make the local choices globally coherent. Distant components cannot cancel a local obstruction: bridge extraction is analytic, its full fibre is (4.2)–(4.3), and every finite algebraic certificate is multihomogeneous under that action. $\square$

Lemma 8.2 (ordinary-triangle contextual sufficiency)

If $N \equiv_{\Delta} N'$, then $N \bowtie_+ N'$.

Proof. A labelled mixed-graph isomorphism transports a physical parameter section. For an ordinary triangle, let $\tau_i : \Theta_i \rightarrow Q$, $i = 1, 2, 3$, be its three normalized orientation maps. At the certified common strict continuous-time point, all three have rank nine, the dimension of the normalized three-boundary K2P tensor space. They therefore cover one common open set $U \subset Q$ and have physical analytic sections.

For an embedded triangle, write the identical external contraction as

$$\mathcal{H} : Q \times C \longrightarrow Y.$$

Choose (q, c) in the nonempty open subset of $U \times C$ where $D\mathcal{H}$ has generic maximal rank R . Since every $\tau_i \times \text{id}_C$ is a submersion,

$$\text{rank } D(\mathcal{H} \circ (\tau_i \times \text{id}_C)) = R$$

for every orientation. Thus $\mathcal{H}(U \times C)$ supplies the same full-dimensional regular contextual germ, even when the three arms reconnect inside a level-2 blob.

For several ordinary-triangle factors, replace Q by the finite product $\prod_j Q_j$, replace U by $\prod_j U_j$, and take the product of the orientation submersions. The same generic-context rank argument applies once, so no induction assumes that a previously chosen context remains generic after the next redirection.

It remains to glue the incidence gauges physically. For the two local sections let $A_e^{(k)}, B_e^{(k)} > 0$, $k = 1, 2$, be their paired- and singleton-sector incidence products. Choose $s_e > 0$ so small that $s_e/A_e^{(k)} < 1/2$ for both k , and then choose $g_e > 0$ so small that $g_e/B_e^{(k)} < 1$ for both. The original and both transformed pairs lie in $\mathcal{D}_+$, since each transformed s -coordinate is below $1/2$. Shrink the germs uniformly. The incidence factors cancel termwise in the global contraction, and Lemma 4.3 gives the full-rank global product germ. $\square$

Theorem 8.3 (K2P-SAME)

For every finite labelled leaf set X and every pair N, N' of binary standard semi-directed strongly tree-child level-2 networks on X , with all edge pairs in $\mathcal{D}_+$ and all inheritance probabilities in $(0, 1)$,

$$\boxed{N \preceq_+ N' \iff N \equiv_\Delta N' \iff N \bowtie_+ N'.} \quad (8.1)$$

In particular, directed regular-germ containment is symmetric within this class: there is no proper one-sided containment. The theorem does not assert equality of complete stochastic images.

Proof. Lemma 8.1 proves necessity and Lemma 8.2 proves sufficiency. A common full-dimensional germ with physical sections gives containment in both directions by definition. $\square$

9. Generic identifiability

Theorem 9.1 (generic structural identifiability)

Fix a binary standard semi-directed strongly tree-child level-2 network N on X . There exists a proper Zariski closed subset

$$E_N \subsetneq \overline{\mathcal{M}_+(N)}^c \quad (9.1)$$

such that every exact physical tensor $q \in \mathcal{M}_+(N) \setminus E_N$ determines the labelled standard semi-directed topology of N uniquely modulo $\equiv_\Delta$.

Proof. For fixed X , only finitely many labelled standard strong level-2 topologies occur. The complex image closure of a fixed parameterization is irreducible. Inside it, take the finite union of the singular locus, the closures of source rank-drop images, the closures of intersections with each non-equivalent topology, and the zero sets of the finitely many nonidentically-zero anchors, rank minors, and certificate pullbacks used by reconstruction.

Every member is proper. The only nonformal point is a competing physical intersection. Stratify the source, target, and incidence set

$$Z_{N'} = \{(q, \theta') : q = \Phi_{N'}(\theta')\} \quad (9.2)$$

into finitely many constant-rank semialgebraic strata. If such an intersection were Zariski dense in the irreducible source closure, it would have full real semialgebraic dimension. Some regular source stratum would then contain an open germ, and some stratum of (9.2) would project onto that germ with full rank. The constant-rank theorem supplies a physical analytic target section after shrinking. This is $N \preceq_+ N'$, contradicting Theorem 8.3 for a non-equivalent N' . $\square$

The quantifier is “for each fixed N , outside a proper exceptional set.” It is not pointwise identifiability at every parameter, and it does not identify individual bridge parameters within the incidence gauge fibre.

10. Exact reconstruction

Theorem 10.1 (finite exact reconstruction)

There is a terminating exact procedure which, given a tensor $q \in \mathcal{M}_+(N) \setminus E_N$ for some binary standard semi-directed strongly tree-child level-2 network N , returns the unique structural class $[N]_\Delta$.

Procedure and proof.

1. Fourier-transform the exact pattern tensor.
2. Evaluate the pointwise quartet sign deck and reconstruct the labelled tree of blobs.
3. Evaluate the whole-map $\mathcal{T}_i$ deck and decorate every degree-three node.
4. Factor the positive bridge blocks in the paired and singleton sectors and apply the analytic incidence normalizers.
5. Evaluate the bounded local certificate deck and select a rigid support.
6. Follow only exact fixed-full restoration records and their stored transports.

7. Apply the frozen one-/two-port probe ledger to recover every boundary segment and the complete label order on each segment.
8. Assemble the coherent labelled mixed graph, quotient the ordinary triangle choices, and return a canonical representative of $[N]_{\Delta}$.

All decks and forests are finite; the restoration depth is at most two and the word reconstruction terminates. The largest bounded restriction in the input contract uses at most nine physical ports, so a direct implementation uses at most $O(|X|^9)$ bounded restrictions beyond reading and transforming the explicit $4^{|X|}$ -entry tensor. No numerical stability or bit-complexity bound is asserted.

The output is a structural triangle class. Other redirected representatives may be listed as structural alternatives, but deciding whether the particular input tensor belongs to any chosen representative requires its own exact semialgebraic membership test. The theorem does not claim otherwise. $\square$

11. Strict continuous-time corollary

Let

$$\mathcal{D}_{\text{CT}} = \{(s, g) : 0 < s < 1, s^2 < g < 1\}. \quad (11.1)$$

This is a nonempty Euclidean-open subset of $\mathcal{D}_+$. Define $\Theta_{\text{CT}}(N)$, $\preceq_{\text{CT}}$, and $\bowtie_{\text{CT}}$ by replacing every edge domain with (11.1).

Theorem 11.1 (continuous-time transfer)

The equivalence (8.1), the generic identifiability statement of Theorem 9.1, and the structural reconstruction of Theorem 10.1 all hold with $+$ replaced by CT.

Proof. Necessity follows because (11.1) is open inside $\mathcal{D}_+$: a continuous-time source germ and target section are also a $\mathcal{D}_+$ germ and section. For sufficiency, positive power roots give continuous-time serial subdivisions and marginal sections. The triangle common rank-nine witness lies strictly in (11.1). For two incidence sections with products $A^{(k)}, B^{(k)} > 0$, choose $s > 0$ sufficiently small and then choose g in the nonempty interval

$$\max \left\{ 1, \frac{B^{(1)}}{(A^{(1)})^2}, \frac{B^{(2)}}{(A^{(2)})^2} \right\} s^2 < g < \min\{1, B^{(1)}, B^{(2)}\}. \quad (11.2)$$

Then the original and both gauge-transformed bridge pairs satisfy (11.1). Finally, every nonzero polynomial witness remains nonzero on a dense open part of the continuous-time parameter space because that space is Euclidean open. The semialgebraic genericity and reconstruction arguments therefore restrict unchanged. $\square$

No boundary case $g = s^2$, $s = 0$, $g = 0$, or inheritance probability in $\{0, 1\}$ is included.

12. Sharpness of strong tree-childness

The following theorem also has a proof independent of the finite probe gate.

Theorem 12.1 (weak-class $4n - 3$ ambiguity)

For every integer $n \geq 3$, there exist two binary level-2 semi-directed networks on the same n -leaf labelled set which are weakly tree-child but not strongly tree-child, are neither labelled-isomorphic nor ordinary-triangle-equivalent, and whose strict continuous-time K2P images contain a common full-dimensional regular analytic germ of dimension

$$4n - 3. \quad (12.1)$$

Proof. On three leaves, use the following two rooted presentations and then suppress the root while retaining reticulation arrowheads. The first has arcs

$$rS, rL_0, SU, SV, UX, VZ, ZX, UV, ZL_1, XL_2,$$

with reticulations V, X . The second has arcs

$$rS, rL_0, SU, SX_0, VX_0, UX_1, VX_1, UV, X_0L_1, X_1L_2,$$

with reticulations X_0, X_1 . They are respectively the certified θ_0 factor with an ordinary boundary on its $V \rightarrow X$ segment and the certified bare θ_3 factor. Complete root-insertion enumeration gives respectively

$$(\# \text{ admissible rootings}, \# \text{ tree-child rootings}) = (5, 2), \quad (7, 2).$$

Thus both are weak but not strong. Exact labelled mixed-graph comparison excludes both isomorphism and ordinary-triangle equivalence.

Put $\delta = 2^{-30}$. On the first network, assign $(s, g) = (1/7, 1/7)$ to its seven internal edge classes, inheritance parameters $(15996/16339, 1/8)$, and pendant pairs $(s, g) = (x, x)$ at leaves 0, 1, 2, where

$$x = \frac{86779}{80}\delta, \quad \frac{320}{253}\delta, \quad \frac{114373}{20240}\delta,$$

respectively. On the second network, assign $(1/4, 1/4)$ to every internal edge class, inheritance parameters $(1/2, 1/6)$, and pendant values

$$\frac{16}{3}\delta, \quad \frac{32}{9}\delta, \quad \frac{96}{5}\delta.$$

Every pair is strict continuous-time. At these rational parameters, the two maps give the same tensor: the six two-nonzero-character orbit coordinates equal δ^2 , the three all-nonzero orbit coordinates equal $\frac{4}{5}\delta^3$. Exact nonzero rational 9×9 minors give rank nine on both sides, the full normalized three-leaf dimension. Hence their images share a regular nine-dimensional germ.

Replace the same labelled leaf on both sides by the same labelled cherry. If its two edge pairs are (u_s, u_g) and (v_s, v_g) , the four local observables

$$R_s = \frac{u_s}{v_s}, \quad P_s = u_s v_s, \quad R_g = \frac{u_g}{v_g}, \quad P_g = u_g v_g$$

have Jacobian determinant

$$\frac{4u_s u_g}{v_s v_g} \neq 0. \tag{12.2}$$

The audited physical witness $(u_s, u_g) = (2/5, 4/9)$, $(v_s, v_g) = (3/7, 5/11)$ lies strictly in (11.1), and (12.2) equals $2464/675$. Thus a cherry adds at least four local dimensions. Conversely the extended tensor factors through the old tensor and these four edge coordinates, so it adds at most four. Iteration gives

$$9 + 4(n - 3) = 4n - 3.$$

Pruning the labelled cherries recovers the base pair. It preserves the existence of both a tree-child and a non-tree-child rooting, and any extended isomorphism or triangle equivalence would prune to one for the base pair. All properties therefore persist. $\square$

The theorem proves that the word “strongly” in Theorem 8.3 cannot be weakened to “weakly,” already within strict continuous time. It does not claim that every weak network is ambiguous.

Appendix A. Frozen evidence manifest

The following hashes are promotion inputs. File hashes are SHA-256 hashes of the exact bytes; payload hashes use the canonical semantic-payload convention of the corresponding package.

Layer	File SHA-256	Payload SHA-256
Domain, subdivision, rooting	4e38beb68062deae8f83cd265daacbef8c5d 3f6d73ce25ef47a54828b658d450	01d03c01482ba1f4f0e43d03c3defbfff35bc 10a97f8b19412ce96e4ce8025328
Two-sector bridge and marginal	9231a7b78c13e54b745eba68926276a6551c 6c3512d6a85746baba6613c1aacf	5abc19f857a02c712d1386b53bf1ecea18ec 31db852cb31b24ea7dde688630ee
Independent component-scale audit	d9b19cde1e9d0544d8c1674ace662b050301 313858c1b9bdc857af29c64804d6	ce8f4e6860675e36238b8351458875bd46de 0507df34f6729c34b20170e02acd
Raw four-port composite ledger	431dac8898ad2a724d12c200687de1b37772 3e302214a79a11a03524a4084b96	9cbb689ab060d4ead06ab5bc995343267ad4 abab7cd62aeca5b4cf39046e0190 (summary)
Raw four-port independent replay	3e067178ca56f94672151dcf42d48e3b05b5 cec411cd709c8b1ae00faf20b620	1a51d5ff1ab6b00fdb16259ac31a457d5d84 fe9c272dbe1ea1c2ba70795e4bbe
Raw four-port terminal registry	0a1818655429d60660c1ed87f3f3be412701f 386b081562b3a4caa54079069f1d	30053d096e140becdf32d3f26cb3634e82fc d5461f89be24f5b83cee344b5b81
Raw four-port mutation report	2e5f23694ca89b1a021dfc58af829b6596cd 497ec7469ca2f038095699b1e672	63eca71e79f2447ec8bb1b3b84396f9fb63a dc9138adb4ab08640ed4a52c341
Five-port θ_2 composite ledger	4cbd7b774adccaafc81338ce9093e33f4abc ae8d75664c9d4c9ecc582a80cc58	6109900c2aa380cdfc0b75fa3a12f283e8d 3a35373e9498a2f976ba7b2c6059 (summary)
Five-port θ_2 independent replay	58c37895cc896c69ba1c25db1476f83cd22d d2e2986535b358222698de94cc42	0e80f8c42cdaef062cc335c871b4dadd9e8 5592c3fad108903c634f448218a4
Five-port θ_2 mutation report	8f1915f313ebcd9012ea0b8448165b7733ca e370e709dece0ad2c662f46b7fbe	18eae9823eaaf3dbf726e6cf60e30abeb2 2aff0e404a06f1f36ecce506e536
Corrected 997-parent restoration forest	43bd2be5e7626a954fc4fa4cf45e8d0e6483 c947ddc9cba80f2b1a13351bc3a8	0a3df52751ba38d7e6d4d118ee7068a98b7b e7897d0aa732e96a74d7523a88bf
Restoration independent replay	24fa2e61f60610a8b24c4107ec7f866278f0 cc671ca203d7aaa40a37bea291dd	36c89ff9729e049a374a9fead8488f7a90e6 2c617d17e242aca5d340faeb164a
Authoritative cycle promotion	b560fdf0545c36d576a4cdf24af9984f6f7 231180f20f6927121a57bf816a7a	df5e3966822af65e2341660bf3f607ff3635 d69d3e5a89854faef308727f2f1
Ordinary-triangle common germ	b81a6cf8da1380f6a682ba6042f6f429ce5d 6a47ba0cf62e9c9d8de1b4158885	6fd43ae6d38629277c047d3888e970cdab51 f4805dce36d71b2430095c1e1aa6
Frozen probe-input contract	7f686ae99dd5e6dafc1c04396b711d294a0b ddd6a25574f9ea809b831ad7b377	579919ca13204ddf959b3a159e4849b69c05 ac87861eba2221659ec45bd73f38
Independent probe-input replay	54de1bef73e76fc82132ef3f0250a0579ea4 01274a302ed4d8fbd015c9e8a053	96d14bae9b20646abfe64b85a7ac0f613771 82f75479031f621ea0dbe2096fce
Corrected full probe primary	6c228e0d495fbc68b1df4ddb571558425a35 e081cc22b954f232f8a3458991d3	674853fa730c4f54b9ba264d539a51591c8b 926ad444195e68df086c26f83825
Independent primitive/graph/full-map probe audit	879a397a165e19c162561435e7996a5de5eb df5e9a7dc38af61a13cc6aa36410	65160636abfa33de47136a222081ac70bd7b 6fae0e029b7a7c379e2d8653df74
Independent probe mutation report	b0df0584163150c9a823b4e364b8ee46c196 ae8abb28fdca4d3d5893a97bf7ea7	58006ed7b6677c055b5cdd7249857dc2f752 fb3db9cfbcc5bbe5e0a26e31875f
Independent structural/algebra probe replay	9b0e45ea62ed30b3502e16677a97a41b9993 6b2f9eb8cf3b97b2156ee2eb5795	b5be07041c7e52cce4828982f295032894c3 51f41a97cd85f8c753709fe1a8d2
Independent site-transport partition replay	017068d960e0bfc3d5e44b79b5acc4a8e931 95c37d205e8c43321c23a0b77e61	a1f80d5ed090a1c6e31067123f0739392384 d5124e75296cc76035430710497c
Weak-sharpness primary	e66c78a0aeab990b4dc448f4f064b37e1e15 ecbff75a5f472bf116d4464378bd	dfecd30ea217810a902add48350025e5f00d fa1255718783df790a9c7e1a5182
Weak-sharpness independent audit	cf4d8d3a2ebc7431d141cac6ebe943e25730e b086fbc84b52833a40bee40a5d52	848cc69e28e3c3db8bc1ab7bbad82b0c3e240 354079e95965b433c423edc2d8c5

All rows required by Gate G_{probe} are present. The promotion guard verifies the primary, independent graph replay, mutation report, and six raw probe ledgers byte-for-byte in addition to the 23 earlier frozen inputs.

Appendix B. Scope exclusions

The main classification does not include stochastic boundary points, singular edge eigenvalues, inheritance probabilities 0 or 1, nonbinary networks, networks of level greater than two, or networks that are merely weakly tree-child. It classifies regular analytic containment germs and generic structural recovery, not full-image equality, universal parameter identifiability, noisy-data estimation, numerical conditioning, or model selection from finite samples.

Reference

The topology-only primitive, repair, and displayed-quartet results reused in the graph layer are from A. K. Englander, M. Frohn, E. Gross, N. Holtgreffe, L. van Iersel, M. Jones, and S. Sullivant, *Identifiability of Phylogenetic Level-2 Networks under the Jukes–Cantor Model*, bioRxiv 2025.04.18.649493, version 4 (2026-07-04), especially Propositions 2.9–2.10, Theorem 2.11, and Corollary 2.12. Only topology-level statements are reused; all algebraic claims in this manuscript are K2P claims certified by the artifacts above.